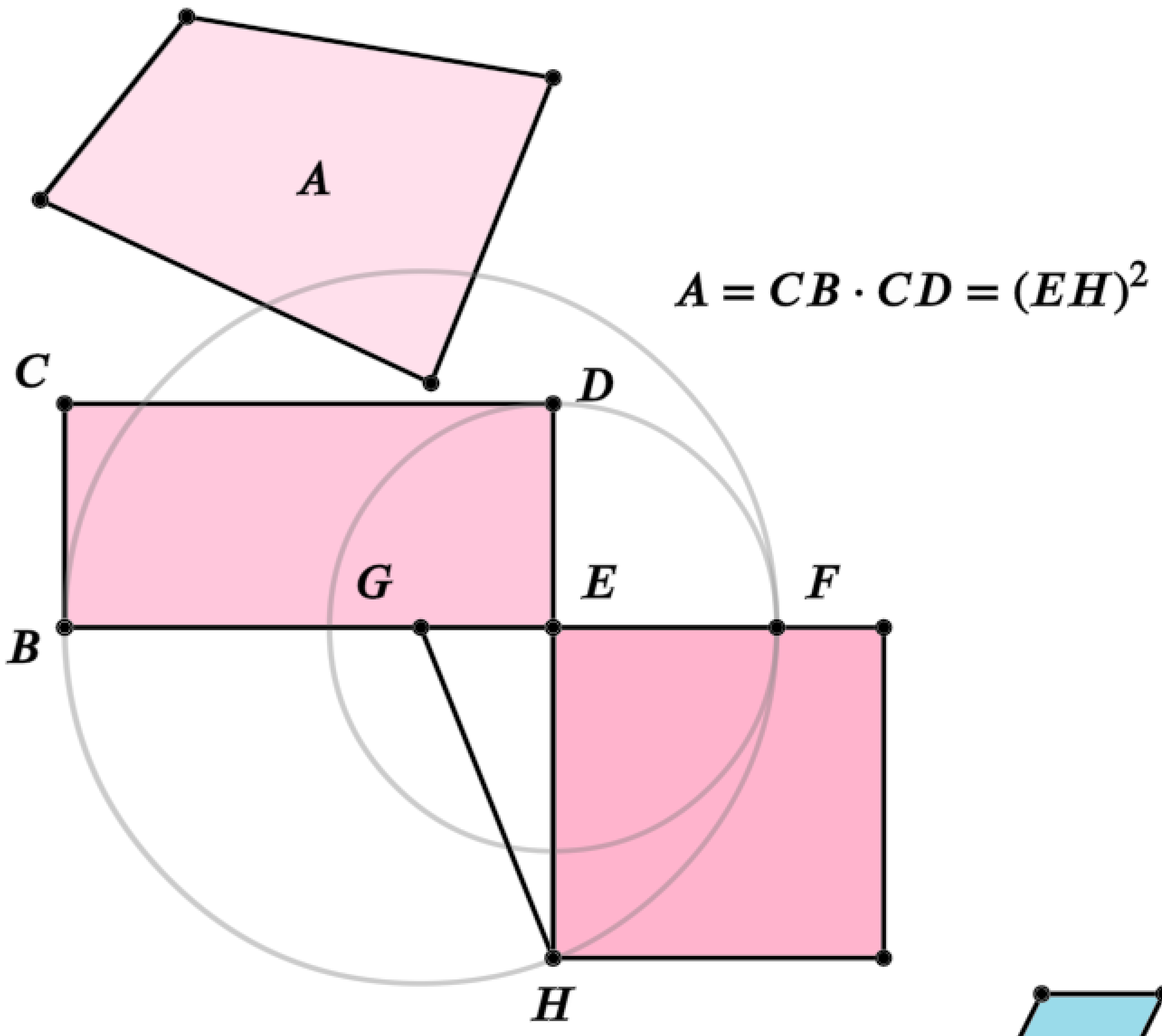
Euclid's Elements

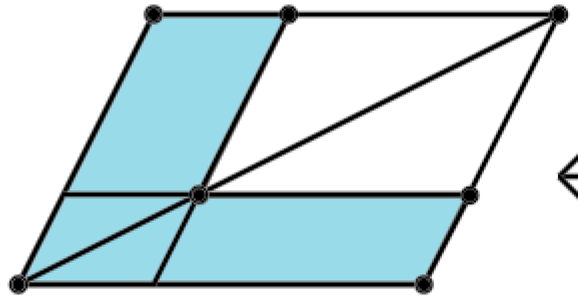
Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)

Definitions:



Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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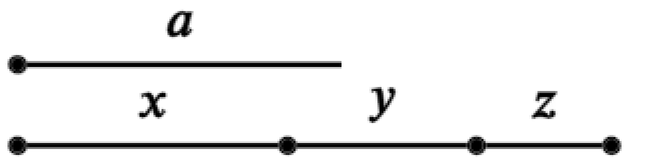
1. 

$a\,(x+y+z) = ax + ay + az$

2. 

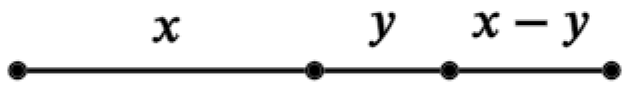
$(x+y)^2 = x\,(x+y) + y\,(x+y)$

3. 

$y\,(x+y) = yx + y^2$

4. 

$(x+y)^2 = x^2 + y^2 + 2xy$

5. 

$(x+y)(x-y) + y^2 = x^2$

6. 

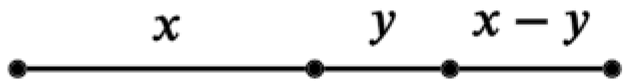
$y\,(2x+y) + x^2 = (x+y)^2$

7. 

$(x+y)^2 + y^2 = x^2 + 2y\,(x+y)$

8. 

$(x+2y)^2 = 4y\,(x+y) + x^2$

9. 

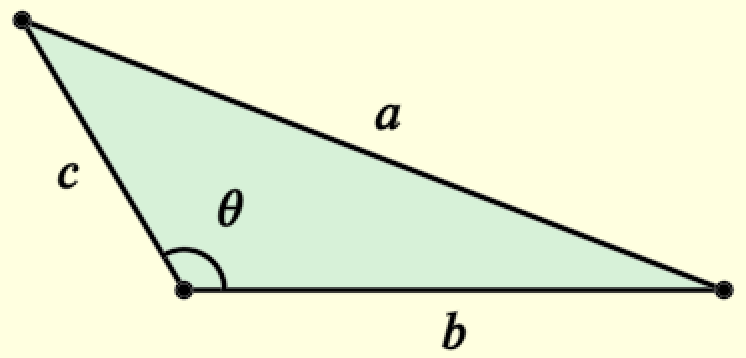
$(x+y)^2 + (x-y)^2 = 2\,(x^2+y^2)$

10. 

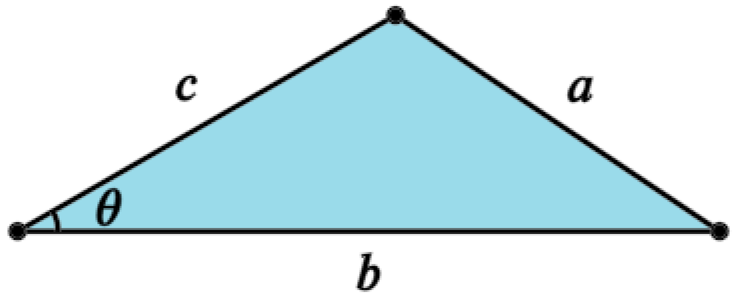
$(2x+y)^2 + y^2 = 2\,(\,x^2 + (x+y)^2\,)$

11. 

Find H such that: $y\,(x+y) = x^2$

12. 

Cosine Law, $a^2 = b^2 + c^2 - 2bc\cos\theta$

13. 

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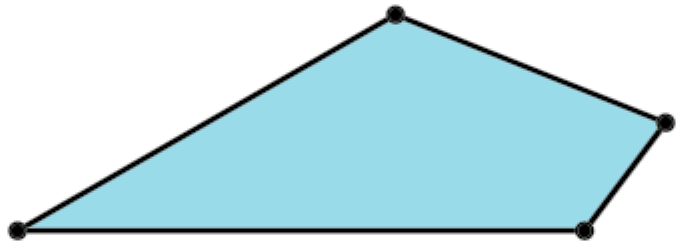
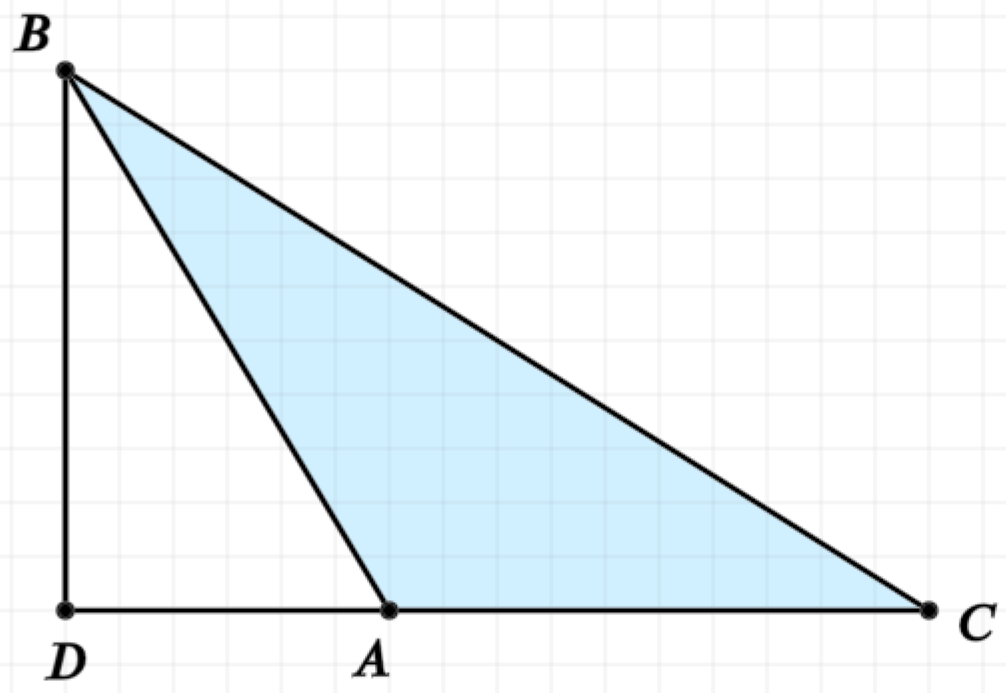
14. 

Construct a square equal to the polygon

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



In other words

Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

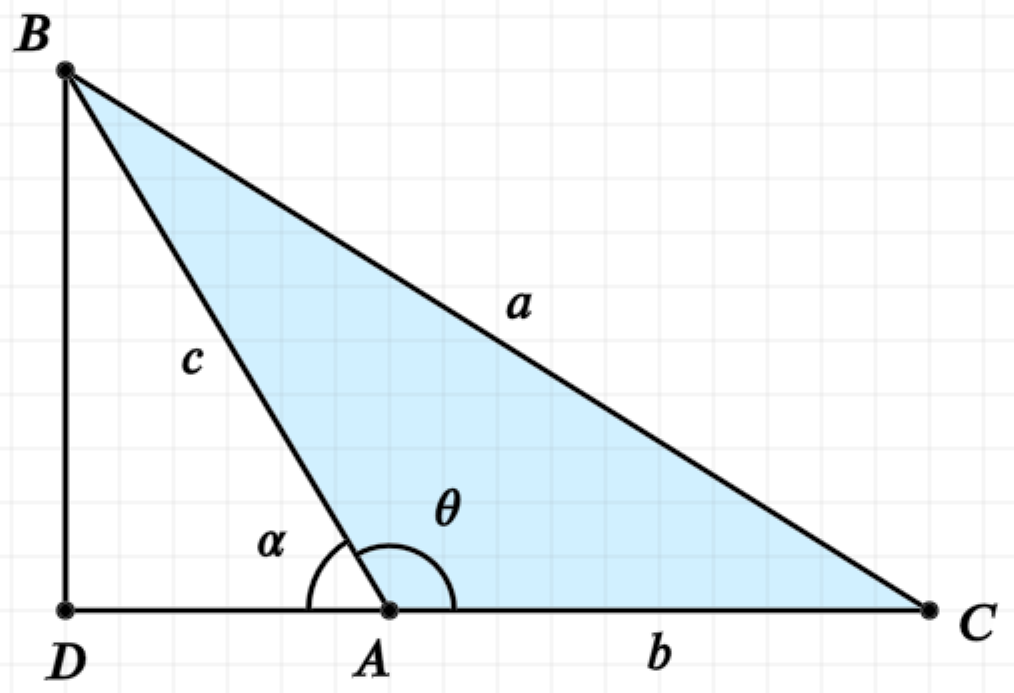
The square of BC equals the sum of the squares of AB and AC, plus...

... twice the rectangle formed by base AC and the extension of the base, AD, where AD is the length from the base to the perpendicular from B to the base

$$BC^2 = AB^2 + AC^2 + 2 \cdot AD \cdot AC$$

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



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Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

The square of BC equals the sum of the squares of AB and AC, plus...

... twice the rectangle formed by base AC and the extension of the base, AD, where AD is the length from the base to the perpendicular from B to the base

$$BC^2 = AB^2 + AC^2 + 2 \cdot AD \cdot AC$$

Or... the cosine law

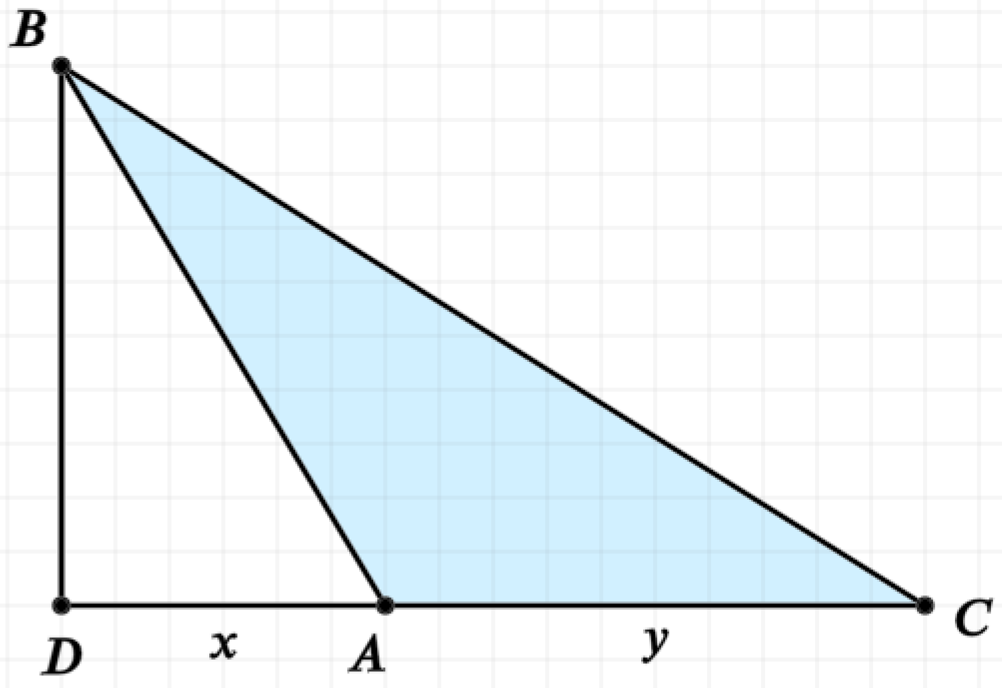
$$BC = a, \quad AB = c, \quad AC = b$$

$$AD = c\cos\alpha = -c\cos\theta$$

$$a^2 = c^2 + b^2 - 2bc\cos\theta$$

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



$$DC^2 = DA^2 + AC^2 + 2 \cdot DA \cdot AC$$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

In other words

Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

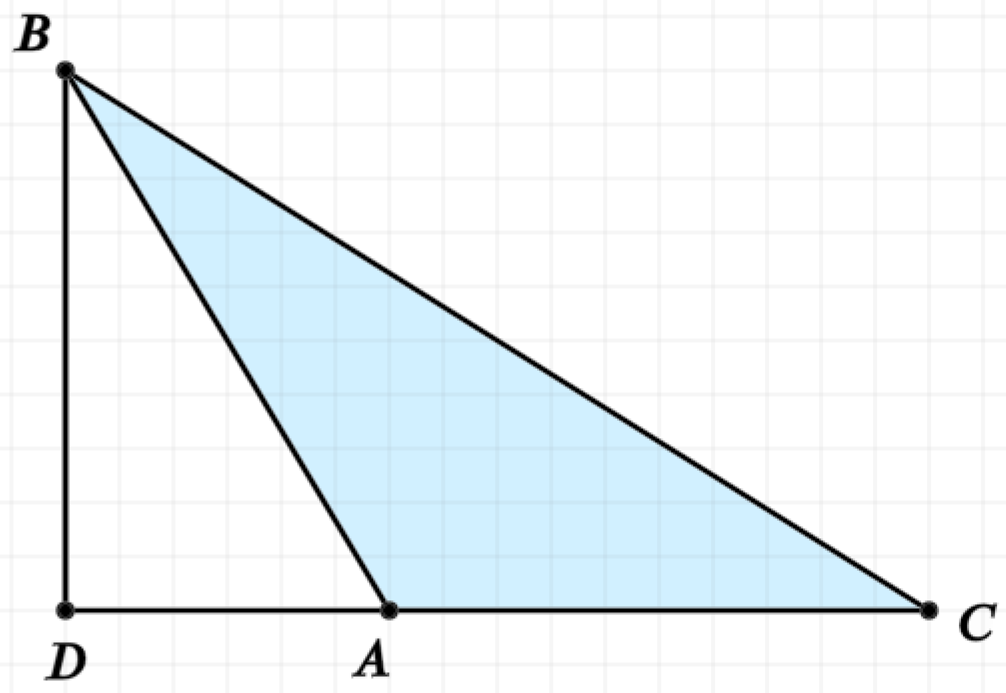
$$BC^2 = AB^2 + AC^2 = 2 \cdot AD \cdot AC$$

Proof

Since the line DC is cut at point A, then we know that the square of DC is equal to the squares of DA and AC plus twice the rectangle formed by DA and AC (II.4)

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



$$DC^2 = DA^2 + AC^2 + 2 \cdot DA \cdot AC$$

$$(DC^2 + DB^2) = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

In other words

Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

$$BC^2 = AB^2 + AC^2 = 2 \cdot AD \cdot AC$$

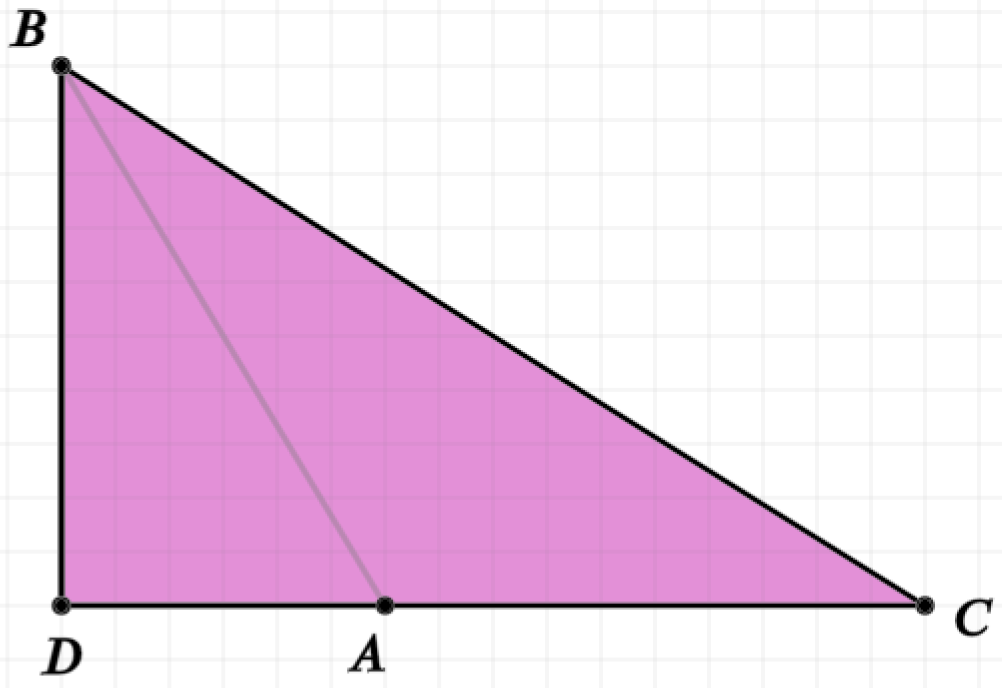
Proof

Since the line DC is cut at point A, then we know that the square of DC is equal to the squares of DA and AC plus twice the rectangle formed by DA and AC (II.4)

Add the square of DB to both sides of the equality

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



$$DC^2 = DA^2 + AC^2 + 2 \cdot DA \cdot AC$$

$$(DC^2 + DB^2) = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

$$DC^2 + DB^2 = BC^2$$

In other words

Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

$$BC^2 = AB^2 + AC^2 = 2 \cdot AD \cdot AC$$

Proof

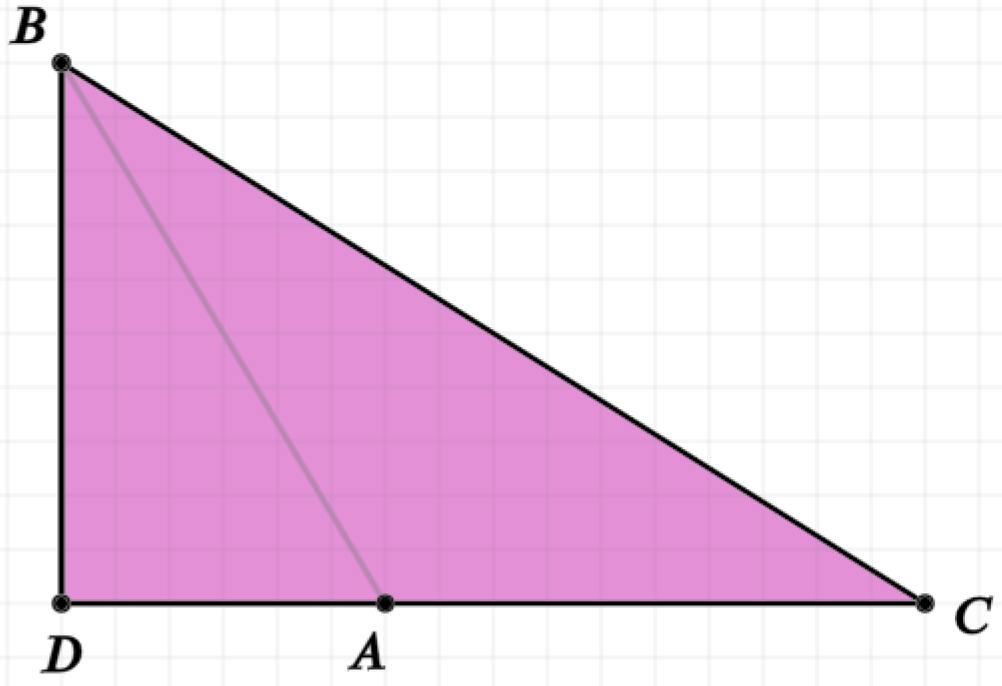
Since the line DC is cut at point A, then we know that the square of DC is equal to the squares of DA and AC plus twice the rectangle formed by DA and AC (II.4)

Add the square of DB to both sides of the equality

The squares of DC and DB equals the square of BC (I.47)

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



$$DC^2 = DA^2 + AC^2 + 2 \cdot DA \cdot AC$$

$$(DC^2 + DB^2) = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

$$DC^2 + DB^2 = BC^2$$

$$BC^2 = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

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Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

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Since the line DC is cut at point A, then we know that the square of DC is equal to the squares of DA and AC plus twice the rectangle formed by DA and AC (II.4)

Add the square of DB to both sides of the equality

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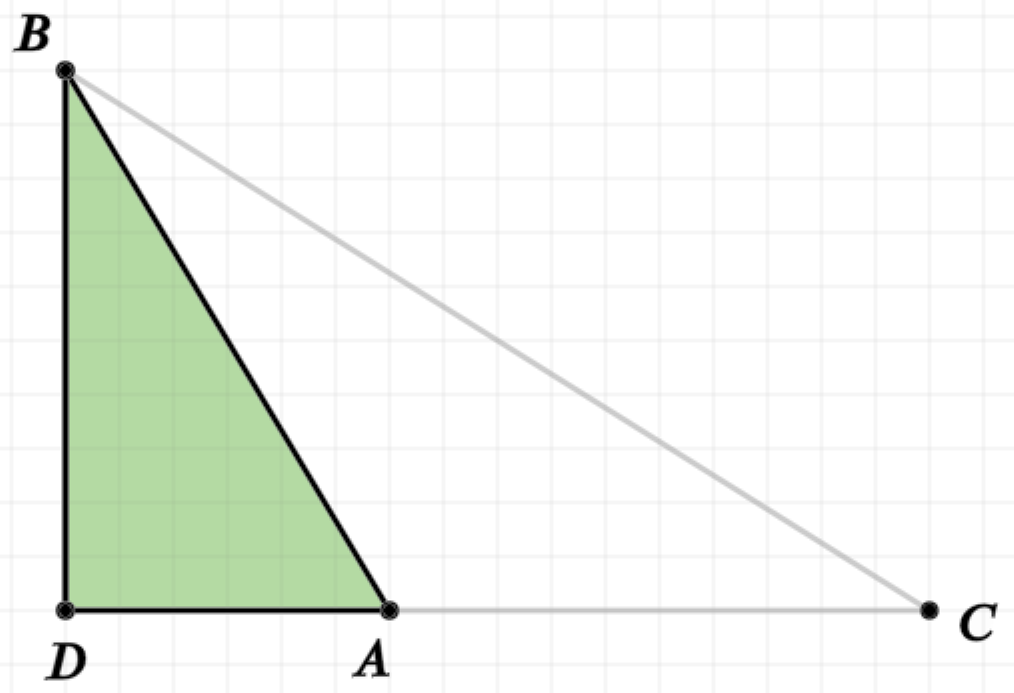
In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



$$DC^2 = DA^2 + AC^2 + 2 \cdot DA \cdot AC$$

$$(DC^2 + DB^2) = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

$$DC^2 + DB^2 = BC^2$$

$$BC^2 = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

$$DA^2 + DB^2 = AB^2$$

In other words

Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

$$BC^2 = AB^2 + AC^2 = 2 \cdot AD \cdot AC$$

Proof

Since the line DC is cut at point A, then we know that the square of DC is equal to the squares of DA and AC plus twice the rectangle formed by DA and AC (II.4)

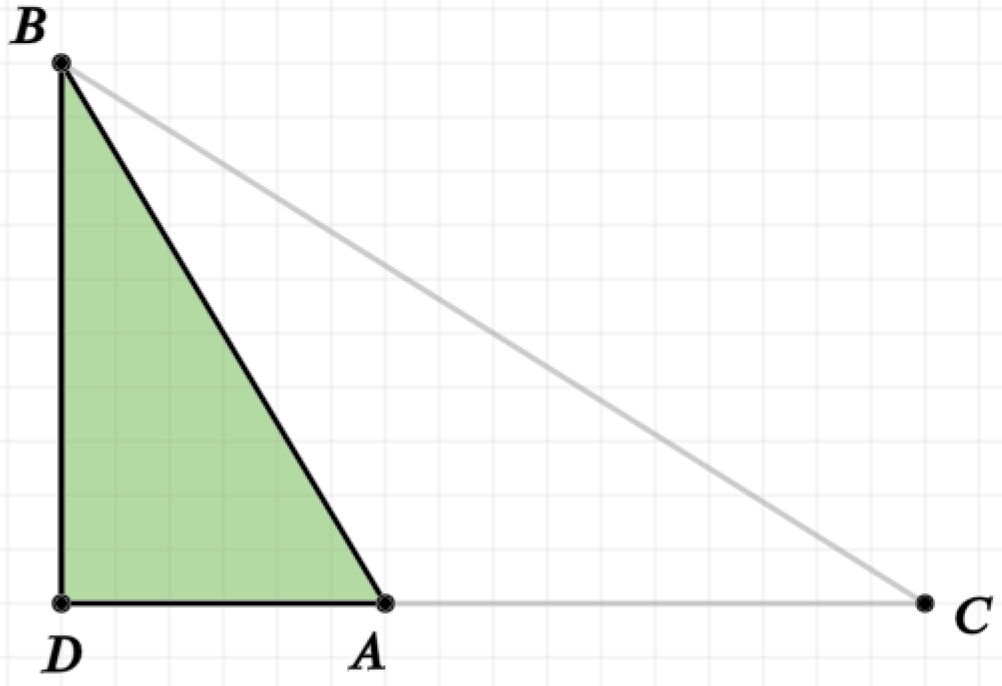
Add the square of DB to both sides of the equality

The squares of DC and DB equals the square of BC (I.47)

The squares of DA and DB equals the square of AB (I.47)

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



$$DC^2 = DA^2 + AC^2 + 2 \cdot DA \cdot AC$$

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$$DC^2 + DB^2 = BC^2$$

$$BC^2 = (DA^2 + DB^2) + AC^2 + 2 \cdot DA \cdot AC$$

$$DA^2 + DB^2 = AB^2$$

$$BC^2 = AB^2 + AC^2 + 2 \cdot DA \cdot AC$$

In other words

Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

$$BC^2 = AB^2 + AC^2 = 2 \cdot AD \cdot AC$$

Proof

Since the line DC is cut at point A, then we know that the square of DC is equal to the squares of DA and AC plus twice the rectangle formed by DA and AC (II.4)

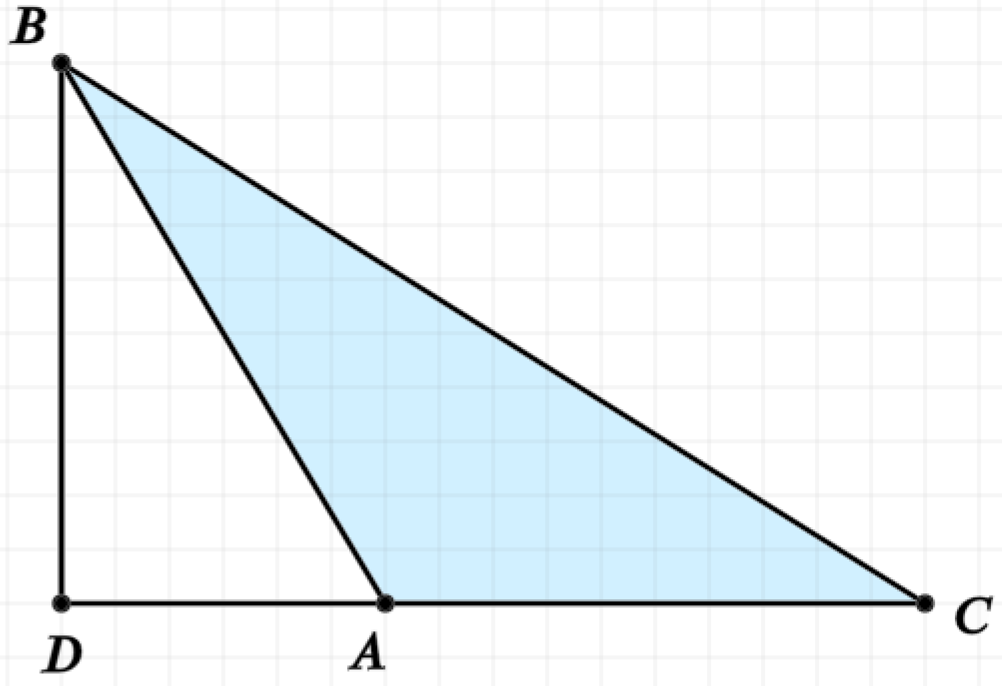
Add the square of DB to both sides of the equality

The squares of DC and DB equals the square of BC (I.47)

The squares of DA and DB equals the square of AB (I.47)

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



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Add the square of DB to both sides of the equality

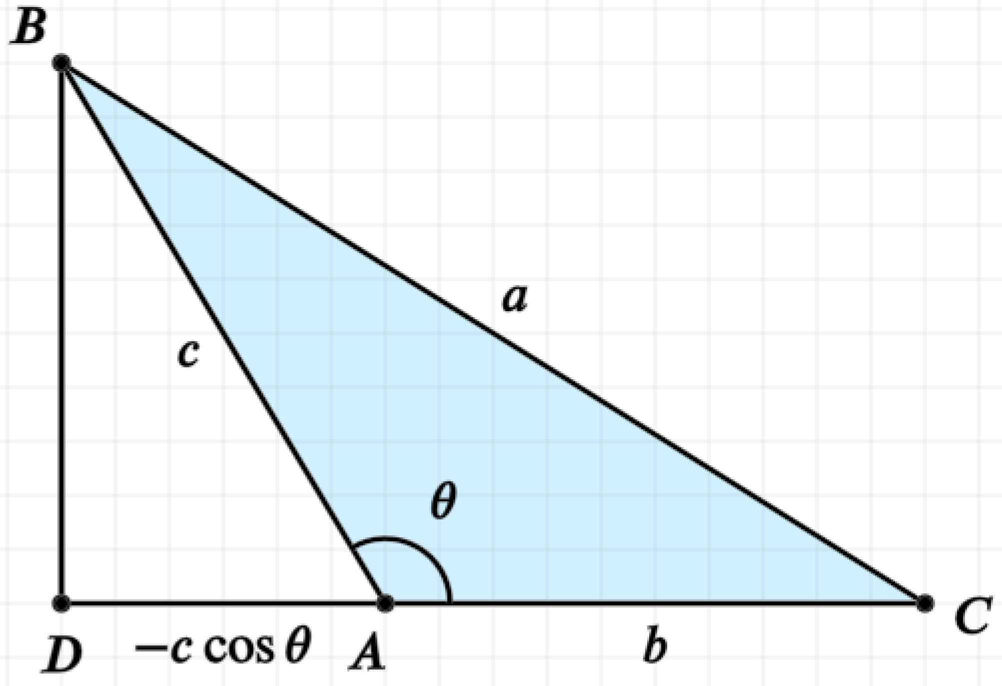
The squares of DC and DB equals the square of BC (I.47)

The squares of DA and DB equals the square of AB (I.47)

Thus the square of BC is equal to the sum of the squares of AB and AC, plus the rectangle formed by DA,AC

Proposition 12 of Book 2

In obtuse-angled triangles the square on the side subtending the obtuse angle is greater than the squares on the sides containing the obtuse angle by twice the rectangle contained by one of the sides about the obtuse angle, namely that on which the perpendicular falls, and the straight line cut off outside by the perpendicular towards the obtuse angle.



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$$DA^2 + DB^2 = AB^2$$

$$BC^2 = AB^2 + AC^2 + 2 \cdot DA \cdot AC$$

$$a^2 = c^2 + b^2 - 2bc\cos\theta$$

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Given an obtuse triangle ABC where the obtuse angle is at point A and the base is AC:

$$BC^2 = AB^2 + AC^2 = 2 \cdot AD \cdot AC$$

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The squares of DA and DB equals the square of AB (I.47)

Thus the square of BC is equal to the sum of the squares of AB and AC, plus the rectangle formed by DA,AC

Content:

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PDFs: https://github.com/sandy-bultena/Euclid/tree/python-manim/Propositions_PDFs

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